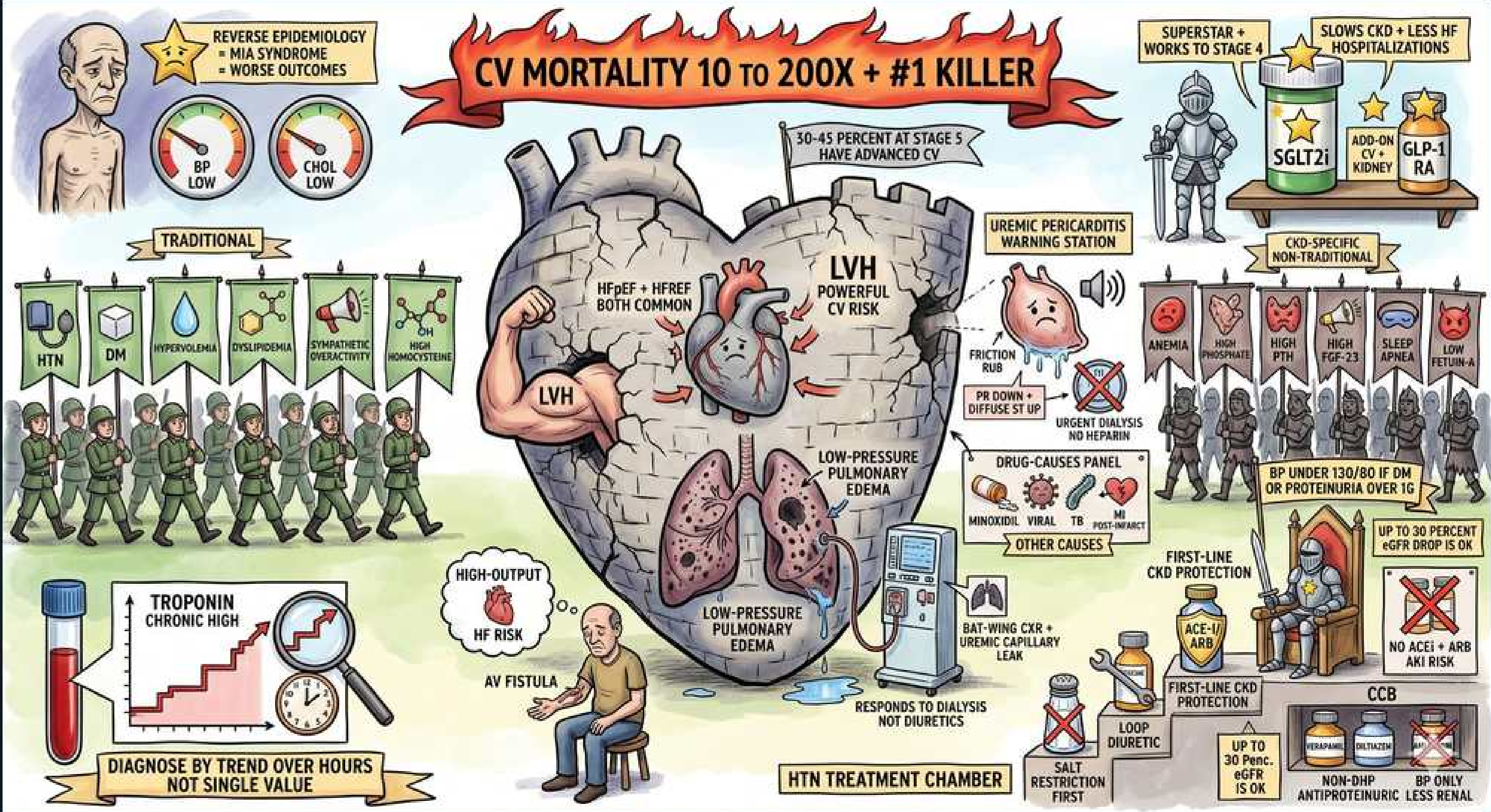
Heart Failure — Cardio-Renal: CKD as CV Risk Multiplier



1. CV mortality 10-200x — #1 killer

WHAT YOU SEE

Top banner: CV mortality 10 to 200X, #1 killer

WHAT IT ENCODES

CKD is a powerful, independent cardiovascular risk multiplier — it raises CV mortality 10- to 200-fold (the relative excess is greatest in the youngest dialysis patients). Cardiovascular DISEASE, not uremia or dialysis complications, is the LEADING cause of death across all stages of CKD. A patient with CKD is far more likely to die of CV disease than to ever reach dialysis. CKD is treated as a coronary-disease risk equivalent.

BOARD TRAP

WRONG answer: 'most patients with CKD ultimately die of progressive kidney failure / uremia.' Discriminator: the #1 cause of death in CKD is cardiovascular disease — most CKD patients die WITH their kidney disease (of CV causes) rather than from end-stage renal failure itself.

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2. 30-45% at stage 5 have advanced CV disease

WHAT YOU SEE

Note: 30-45% at stage 5 advanced CV

WHAT IT ENCODES

By the time CKD reaches stage 5 (eGFR <15 / dialysis), roughly 30–45% of patients already have ADVANCED, established cardiovascular disease — coronary disease, LVH, and HF accumulate silently throughout earlier stages. This underscores starting CV risk reduction EARLY (BP control, statins, RAAS blockade for proteinuria) rather than waiting for symptoms.

BOARD TRAP

WRONG answer: 'cardiovascular screening and prevention can wait until a CKD patient reaches dialysis.' Discriminator: 30–45% already have advanced CV disease by stage 5, so CV risk modification must begin in early CKD — deferring it to dialysis misses the prevention window.

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3. LVH — powerful CV risk

WHAT YOU SEE

Castle heart flexing LVH muscle

WHAT IT ENCODES

Left ventricular hypertrophy is extremely prevalent in CKD (driven by pressure overload from hypertension/arterial stiffness, volume overload from salt/water retention and anemia, and uremic factors) and is one of the strongest INDEPENDENT predictors of CV death. LVH begets diastolic dysfunction, arrhythmia, and sudden cardiac death — the leading mode of CV death in dialysis patients.

BOARD TRAP

WRONG answer: 'LVH in CKD is a benign adaptive change with no prognostic weight.' Discriminator: LVH is a powerful independent predictor of CV mortality and sudden death in CKD — it is a high-risk marker, not a harmless adaptation.

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4. HFrEF + HFpEF both common

WHAT YOU SEE

Banner: HFrEF + HFpEF both common

WHAT IT ENCODES

CKD predisposes to BOTH heart-failure phenotypes: HFpEF (from LVH, hypertension, and arterial stiffness causing diastolic dysfunction) and HFrEF (from accelerated coronary disease/ischemia and chronic volume overload). Diagnosis is harder because natriuretic peptides (BNP/NT-proBNP) are CHRONICALLY elevated in CKD from reduced renal clearance and chronic strain — interpret them against CKD-specific, higher thresholds.

BOARD TRAP

WRONG answer: 'a high BNP/NT-proBNP in a CKD patient reliably confirms acute decompensated heart failure.' Discriminator: natriuretic peptides are chronically elevated in CKD due to reduced clearance — a single elevated value is non-specific; use higher CKD-adjusted cutoffs and the clinical trend/picture.

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5. Low-pressure pulmonary edema

WHAT YOU SEE

Lungs flooding, low-pressure pulm edema

WHAT IT ENCODES

Uremia increases pulmonary capillary PERMEABILITY, producing 'low-pressure' (non-cardiogenic) pulmonary edema — a classic 'bat-wing' central perihilar pattern on CXR — even when left ventricular filling pressures are normal/near-normal. Because the leak is from permeability plus volume overload rather than purely high LV pressure, the definitive treatment is volume removal by DIALYSIS/ultrafiltration.

BOARD TRAP

WRONG answer: 'pulmonary edema in a CKD patient always indicates elevated left ventricular filling pressures and primary cardiac failure.' Discriminator: uremic 'low-pressure' pulmonary edema arises from increased capillary permeability + volume and can occur with normal LV filling pressures — it responds to dialysis, distinguishing it from purely cardiogenic edema.

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6. Responds to dialysis, not diuretics

WHAT YOU SEE

Note: responds to dialysis not diuretics

WHAT IT ENCODES

In advanced CKD/ESRD, the kidneys can no longer respond to loop diuretics (no/low residual function), so uremic low-pressure pulmonary edema and volume overload are treated by DIALYSIS with ultrafiltration, not by escalating furosemide. In earlier CKD with residual function, high-dose loop diuretics still work but need higher doses to reach the tubular site of action.

BOARD TRAP

WRONG answer: 'give escalating IV furosemide to manage flash pulmonary edema in an anuric dialysis patient.' Discriminator: an anuric ESRD patient cannot respond to diuretics — emergent ultrafiltration/dialysis is the treatment for their volume overload and uremic pulmonary edema.

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7. Uremic pericarditis

WHAT YOU SEE

Friction-rub icon, uremic pericarditis station

WHAT IT ENCODES

Uremic pericarditis presents with a pericardial friction rub and chest pain, typically when BUN is very high (often >60 mg/dL); classically the ECG LACKS the diffuse ST elevation seen in viral pericarditis because the inflammation is epicardial/myocardial-sparing. It is an absolute indication to INITIATE or INTENSIFY dialysis, which usually resolves it. Watch for progression to hemorrhagic effusion and tamponade.

BOARD TRAP

WRONG answer: 'treat uremic pericarditis first-line with NSAIDs and colchicine like viral/idiopathic pericarditis.' Discriminator: the primary treatment for uremic pericarditis is intensified DIALYSIS (it reflects under-dialysis); NSAIDs/colchicine are not the answer, and uremic pericarditis classically lacks the diffuse ST elevation of viral pericarditis.

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8. No heparin during dialysis

WHAT YOU SEE

No-heparin warning at pericarditis station

WHAT IT ENCODES

When intensifying dialysis to treat uremic pericarditis, use HEPARIN-FREE (or regional citrate) dialysis. The inflamed pericardium is prone to hemorrhagic effusion, and systemic anticoagulation risks converting it into a life-threatening hemorrhagic effusion/cardiac tamponade.

BOARD TRAP

WRONG answer: 'use standard systemic heparin anticoagulation during dialysis for a patient with uremic pericarditis.' Discriminator: heparin can precipitate hemorrhagic pericardial effusion and tamponade in uremic pericarditis — heparin-free or regional citrate anticoagulation is mandatory.

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9. Troponin — trend over hours

WHAT YOU SEE

Troponin chronically high, trend graph

WHAT IT ENCODES

Cardiac troponin (especially high-sensitivity assays) is CHRONICALLY elevated at baseline in CKD due to reduced renal clearance plus ongoing subclinical myocardial strain/microinfarction. Therefore MI is diagnosed not by a single value above the threshold but by a significant dynamic RISE-AND/OR-FALL (delta) over serial measurements (e.g., 0/1–3 h), interpreted with symptoms and ECG.

BOARD TRAP

WRONG answer: 'a single elevated high-sensitivity troponin in a CKD patient confirms acute myocardial infarction.' Discriminator: baseline troponin is chronically elevated in CKD — diagnosis of MI requires a dynamic delta (rise/fall) on serial draws, not one static elevated value.

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10. MIA paradox — reverse epidemiology

WHAT YOU SEE

Cachectic figure, reverse epidemiology, low BP/chol = worse

WHAT IT ENCODES

The MIA syndrome (Malnutrition–Inflammation–Atherosclerosis/cachexia) drives 'reverse epidemiology' in dialysis patients: factors that are RISK markers in the general population become PROTECTIVE-appearing here — LOW BMI, LOW blood pressure, and LOW cholesterol all predict WORSE survival on dialysis (they mark malnutrition/inflammation/frailty). This is why aggressive statin initiation in prevalent dialysis patients failed to reduce mortality (4D, AURORA trials).

BOARD TRAP

WRONG answer: 'a low cholesterol and low blood pressure in a dialysis patient are reassuring signs of good cardiovascular health.' Discriminator: by reverse epidemiology, low cholesterol/BP/BMI in dialysis patients signal malnutrition-inflammation (MIA) and predict HIGHER mortality — they are bad prognostic signs, not protective.

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11. AV fistula → high-output HF

WHAT YOU SEE

AV fistula arm, high-output HF magnifier

WHAT IT ENCODES

A hemodialysis arteriovenous fistula creates a permanent left-to-right shunt; a high-flow access (especially flow >2 L/min or upper-arm fistulas) can chronically increase venous return and cardiac output enough to precipitate HIGH-OUTPUT heart failure. A bedside clue is the Nicoladoni–Branham sign (manual compression of the fistula slows the heart rate). Treatment is flow reduction (banding) or ligation.

BOARD TRAP

WRONG answer: 'heart failure in a dialysis patient with a large AV fistula must be a low-output ischemic cardiomyopathy.' Discriminator: a high-flow AV fistula can cause HIGH-output HF via shunt physiology — check for the Branham sign (HR drops with fistula compression); banding/ligation can treat it, unlike an ischemic cause.

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12. HTN 3-step: salt restriction first

WHAT YOU SEE

HTN treatment chamber step 1, salt restriction

WHAT IT ENCODES

CKD hypertension is largely volume/salt-dependent, so step 1 is SODIUM RESTRICTION (<2 g/day) plus volume management. BP targets are stringent: generally <130/80 (KDIGO even suggests SBP <120 by standardized measurement in many CKD patients), and tighter control especially with diabetes or proteinuria >1 g/day. Controlling volume and salt also restores responsiveness to antihypertensive drugs.

BOARD TRAP

WRONG answer: 'a BP of 145/90 is an acceptable target in a proteinuric CKD patient.' Discriminator: CKD (especially with proteinuria/DM) requires a tight target ≤130/80 (often lower) plus salt restriction — a 'standard' goal of <140/90 is too lax and accelerates renal and CV decline.

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13. First-line: ACEi/ARB if proteinuria

WHAT YOU SEE

First-line CKD protection, ACEi/ARB

WHAT IT ENCODES

For CKD with proteinuria/albuminuria, an ACE inhibitor OR an ARB is first-line and RENOPROTECTIVE beyond BP lowering — they dilate the efferent arteriole, reducing intraglomerular pressure and proteinuria, slowing CKD progression. Expect (and tolerate) an initial creatinine rise up to ~30% and monitor potassium. Add a loop diuretic for volume. Newer agents (SGLT2 inhibitors, finerenone) add further renal/CV protection in diabetic CKD.

BOARD TRAP

WRONG answer: 'combine an ACE inhibitor WITH an ARB (dual RAAS blockade) to maximize antiproteinuric effect.' Discriminator: dual ACEi+ARB blockade increases hyperkalemia, AKI, and hypotension without outcome benefit (ONTARGET, VA NEPHRON-D) — use only ONE RAAS agent. Also: a creatinine rise up to ~30% is expected and not a reason to stop.

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14. Non-DHP CCB if BP only

WHAT YOU SEE

Non-DHP CCB box, BP only, no proteinuria benefit

WHAT IT ENCODES

Non-dihydropyridine calcium channel blockers (diltiazem, verapamil) lower BP and have a modest antiproteinuric effect, but they LACK the robust renoprotective/antiproteinuric benefit of RAAS blockade and are not first-line for proteinuric CKD. They are useful add-ons (or when RAAS inhibitors are contraindicated/not tolerated). Avoid combining non-DHP CCBs with beta-blockers (risk of bradycardia/heart block).

BOARD TRAP

WRONG answer: 'a non-DHP calcium channel blocker is the preferred first-line antihypertensive for renoprotection in proteinuric CKD.' Discriminator: RAAS blockade (ACEi/ARB) is first-line for proteinuric CKD; non-DHP CCBs control BP but do not match the antiproteinuric/renoprotective benefit — they are adjuncts, not the cornerstone.
